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IEEE Access - Decision on Manuscript ID Access-2026-14298

1 개의 메일

IEEE Access <onbehalfof@manuscriptcentral.com> 2026년 5월 1일 오전 2:09
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30-Apr-2026

Dear Mr. KANG:

I am writing to you regarding manuscript # Access-2026-14298 entitled "Topology-Constrained NeuroB-Rep: Contact-Aware CAD Reconstruction from 3D Point Clouds" which you submitted to IEEE *Access*.

Your article was peer reviewed with interest but has not been recommended for publication in its current form. We strongly encourage you to address the reviewers' concerns, which can be found at the bottom of this letter, and resubmit your article to IEEE *Access* once you have updated it accordingly.

Please note that IEEE *Access* has a binary peer review process. Therefore, to uphold quality to IEEE standards, an article is rejected even if it requires minor edits.

When updating your manuscript, you should elaborate on your points and clarify with references, examples, data, etc. If you disagree with any technical points the reviewers have made, please include your counterarguments in your response to the reviewers (more information detailed below) and work this into the updated manuscript.

Also, note that if a reviewer suggested references, you should only add those that are relevant to your work if you feel they strengthen your article. Recommending references to specific publications is not appropriate for reviewers and you should report excessive cases to ieeeaccessEIC@ieee.org. Authors are not obligated to cite articles that are recommended by the reviewers, and the final decision on the article will not be influenced by whether or not authors cite these suggested references.

IEEE *Access* allows one opportunity to resubmit. If the updated manuscript is determined not to have addressed all of the previous reviewers' concerns, or if the Associate Editor still has substantial technical concerns, the article will be rejected and no further resubmissions will be allowed.

When you are ready to resubmit your updated article, you can do so in the IEEE Author Portal. When you log into the IEEE Author Portal you ill see the title of the rejected article and the option to "Start Resubmission".

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Upon resubmission you will be asked to upload the following 3 files:

1) A document containing your response to reviewers from the previous peer review. The "response to reviewers" document (template attached) should have the following regarding each comment: a) Reviewer's concern, b) your response to the concern, c) your action to remedy the concern. **The document should be uploaded with your manuscript files under "*Author's Response Files*."**

NOTE: Though all attempts are made to use the same reviewers in each round of review, your article may have new reviewers in this round, for a number of reasons. Please address all reviewer comments.

2) Your updated manuscript with all your individual changes highlighted, including grammatical changes (e.g. preferably with the yellow highlight tool within the pdf file). This file should be uploaded with your manuscript files as *"Highlighted PDF."*

3) A clean copy of the final manuscript (without highlighted changes) submitted as a Word or LaTeX file, and as a PDF, both submitted as the "*Main Manuscript*."

****IMPORTANT:** Please see the attached Resubmission Checklist that details all the items listed above. Please utilize this checklist to ensure you have made the necessary edits to your manuscript, and to ensure you have all the necessary files prepared prior to resubmission.

*** **AUTHOR LIST CHANGES:** If your revised manuscript has an updated author list, you will need to submit a formal request to the Editor by completing the attachment labelled ‘Request for Byline Change,’ and uploading it as *'Request for byline change form.'* This should include a DETAILED justification explaining each author’s contribution(s) to the work. You will also need to provide the justification for the author change during the submission process. Change in the author list is considered rare and exceptional, and the decision to allow such changes rests with the Editor. Once the list and order of authors has been established, the list and order of authors should not be altered without permission of all living authors of that article.

We sincerely hope you will update your manuscript and resubmit soon. Please contact me if you have any questions.

Thank you for your interest in IEEE *Access*.

Sincerely,

Dr. Sotirios Goudos
Associate Editor, IEEE *Access*
sgoudo@physics.auth.gr

Reviewers' Comments to Author:

Reviewer: 1

Comments:
The article has significantly improved in this iteration. The reviewer would like to pose additional questions

Comment #1
The authors are kindly requested to
-provide experiments with noise levels in the point cloud
-provide experiments with variable point cloud density
-increase the number of models that they test their approach upon, perhaps by sampling existing industrial CAD models

Comment #2
Figure 6 is showcasing the result of misalignmnet on a complex model, however the same model is reconstructed in Figure 7.
When does this misalignment occur? Testing on additional models and use case would definitetely strengthen the paper. Currently only 3 models are presented

Additional Questions:
Please confirm that you have reviewed all relevant files, including supplementary files and any author response files, which can be found in the "View Author's Response" link above (author responses will only appear for resubmissions): Yes, all files have been reviewed

1) Does the paper contribute to the body of knowledge?: Yes

2) Is the paper technically sound?: Yes

3) Is the subject matter presented in a comprehensive manner?: Yes

4) Are the references provided applicable and sufficient?: No

5) Are there references that are not appropriate for the topic being discussed?: No

5a) If yes, then please indicate which references should be removed.:

Reviewer: 2

Comments:
Thank you for a very thorough and thoughtful revision. Your detailed "Response to Reviewers" document clearly outlined the improvements made to the manuscript.

The addition of Table 7, which provides an apples-to-apples comparison against Point2CAD, significantly strengthens your technical claims and proves the efficacy of your multi-objective topology constraints. Furthermore, restructuring the Related Work section and adding the explicit feature comparisons (Table 1 and 2) greatly improves the paper's readability and contextual placement.

Additional Questions:
Please confirm that you have reviewed all relevant files, including supplementary files and any author response files, which can be found in the "View Author's Response" link above (author responses will only appear for resubmissions): Yes, all files have been reviewed

1) Does the paper contribute to the body of knowledge?: Yes. Automating the conversion of raw 3D scans into precise, watertight, and manufacturable CAD models remains a significant bottleneck in industrial reverse engineering. The proposed Topology-Constrained NeuroB-Rep successfully addresses this by treating topology as a first-class objective. By coupling a hybrid surface model with a contact-aware energy function that penalizes inter-surface gaps and G1 discontinuities, the paper presents a clear advancement over purely distance-based learning pipelines.

2) Is the paper technically sound?: Yes. The methodology is robust and the authors have thoroughly validated their approach. In their revision, the authors added an apples-to-apples quantitative comparison against the Point2CAD baseline under an identical ICP protocol. The results demonstrate that NeuroB-Rep achieves geometric accuracy errors in the 0.01-0.03 mm range, consistently outperforming the baseline with 2x to 3x lower errors. The addition of the PCA normalization ablation (Table 6) further proves the stability of their pipeline.

3) Is the subject matter presented in a comprehensive manner?: Yes. The authors have done an excellent job addressing prior reviewer concerns. The manuscript now clearly delineates its contributions from existing works using detailed stage-by-stage and qualitative comparisons (Tables 1 and 2). Furthermore, the addition of Section VIII-E honestly and comprehensively analyzes the primary failure mode (cross-boundary misassignment due to upstream segmentation) and outlines concrete mitigation strategies.

4) Are the references provided applicable and sufficient?: Yes. The Related Work section has been appropriately restructured to cover point-cloud segmentation, primitive fitting, B-Rep learning, and hybrid reconstruction, properly contextualizing the proposed method.

5) Are there references that are not appropriate for the topic being discussed?: No

5a) If yes, then please indicate which references should be removed.:

If you have any questions, please contact article administrator: Miss Manpreet Kaur m.kaur@ieee.org

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첨부파일 3 개

*** IEEE-Access-Response-to-Reviewers-template-6.26.23.docx**
18K

*** Request-for-Byline-Change--2-.docx**
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*** Resubmission-Checklist-6.26.23.docx**
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